



Customer Shopping Behavior Analysis

Transaction data: 3,900 purchases — discover segments, patterns, product trends, subscriptions.

Dataset Overview

Rows & Columns

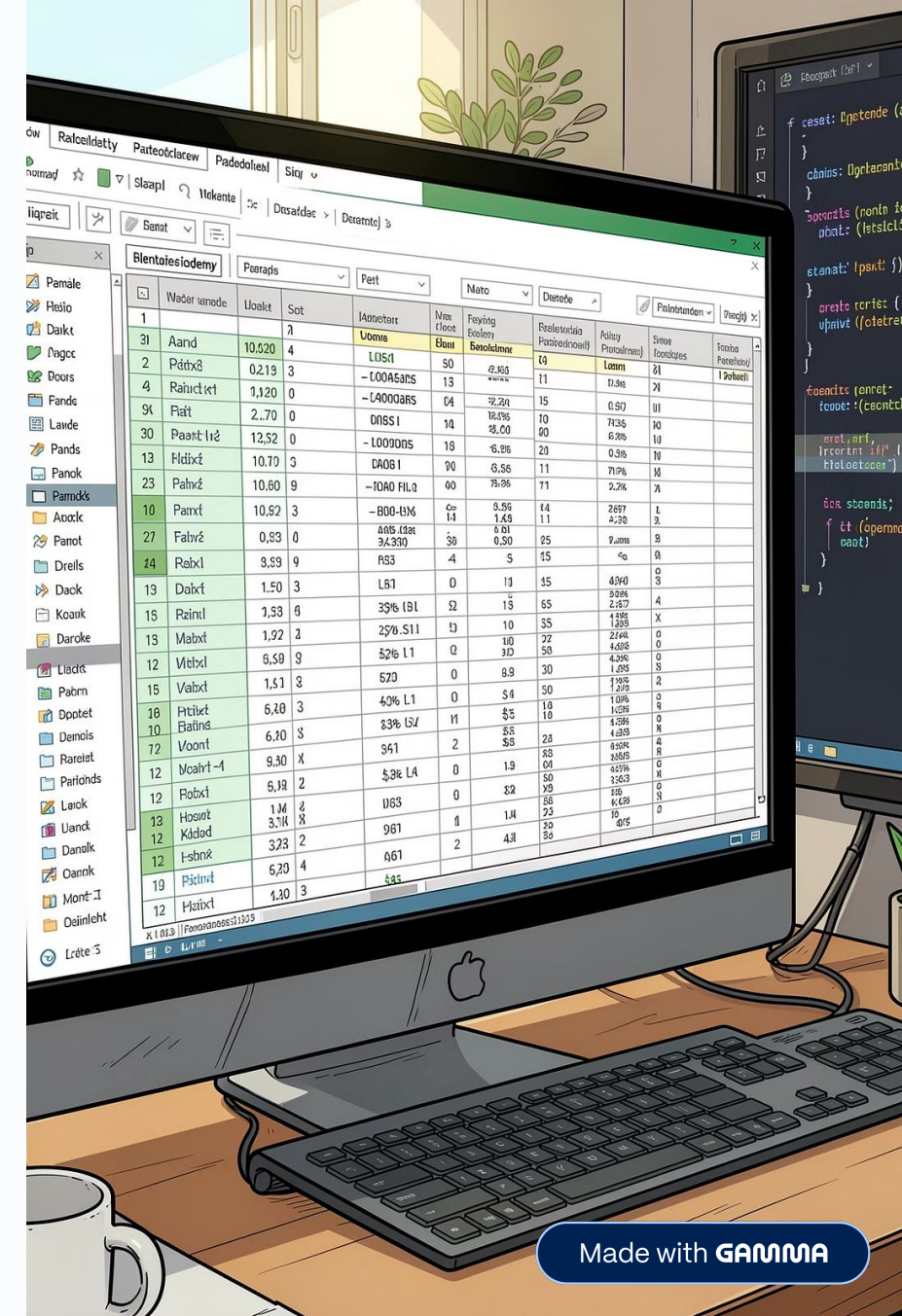
Rows: 3,900 · Columns: 18

Key Features

Demographics, purchase details, behavior, subscription

Missing Data

37 nulls in Review Rating



EDA & Cleaning (Python)

01

Load & Inspect

`df.info(), .describe()`

02

Handle Missing

Imputed Review Rating by product-category median

03

Standardize

Renamed to snake_case

04

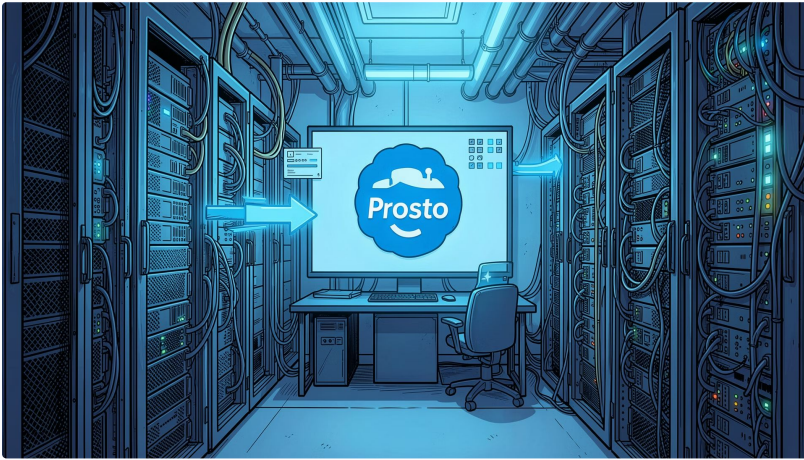
Feature Eng.

`age_group, purchase_frequency_days`

05

DB Load

Exported cleaned DF to PostgreSQL



SQL Analysis Highlights

- Revenue comparisons, discount analysis, top products
- Customer segmentation: New / Returning / Loyal
- Repeat-buyers vs subscriptions

Female
Revenue: 75,191

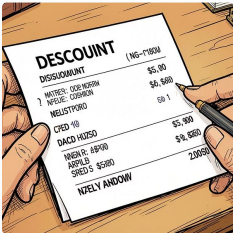
Male
Revenue: 157,890

Revenue: 75,191

Revenue: 157,890



Discounts & High-Spenders



High-Spending Discount Users

839 rows identified — customers using discounts but spending above avg.



Discount-Dependent Products

Top: Hat (50%), Sneakers (49.66%), Coat (49.07%), Sweater (48.17%), Pants (47.37%)

CUANOH4DEBIRES



\$310.00



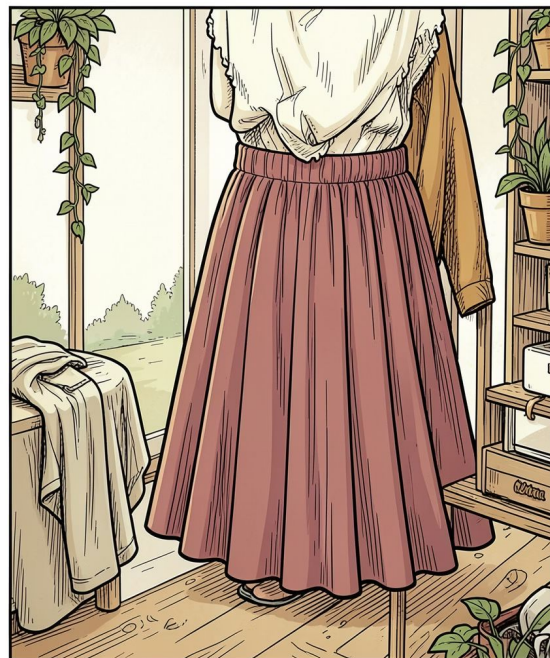
\$56.005



25.0045



\$8100



Top Products & Shipping



Top 5 by Rating

Gloves 3.86 · Sandals 3.84 · Boots 3.82 ·
Hat 3.80 · Skirt 3.78

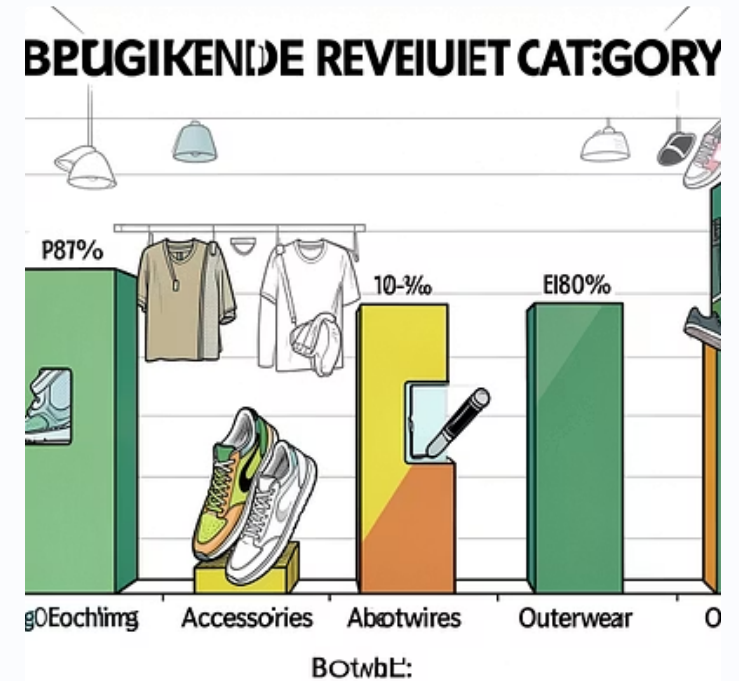
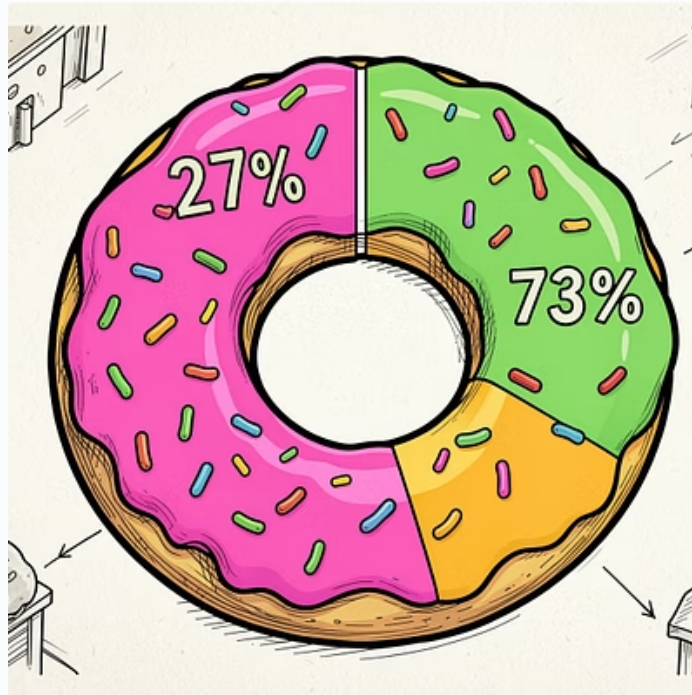


Shipping Comparison

Standard avg \$58.46 · Express avg \$60.48

Power BI Dashboard

Interactive visual summary: subscriptions, revenue by category, sales by category, age-group trends.



Customer Segments & Revenue by Age

Segments

Loyal: 3,116 · Returning: 701 · New: 83

Subscribers

Yes: 1,053 · Avg spend \$59.49 · Revenue 62,645

Non-Subscribers

No: 2,847 · Avg spend \$59.87 · Revenue 170,436

Revenue by Age Group

Young Adult 62,143 · Middle-aged 59,197 · Adult 55,978 · Senior 55,763

Business Recommendations

Boost Subscriptions

Promote exclusive benefits to increase subscriber share.

Loyalty Programs

Reward repeat buyers to grow Loyal segment.

Review Discount Policy

Balance sales uplift vs margins; target discount-dependent SKUs.

Product Positioning

Highlight top-rated & best-selling items in campaigns.

Targeted Marketing

Focus on high-revenue age groups and express-shipping users.